US Stock Market Network Analysis for Crisis Prediction

Matéo Molinaro – Habib Karamoko *30 April 2026*

Abstract

We model the cross-sectional dependence of large-cap U.S. equities as a dynamic weighted graph built from rolling Ledoit–Wolf shrinkage correlation matrices with exponential time-weighting. Graph-level topology statistics and time-series return features are fed into a binary classifier predicting whether the forward 21-day maximum drawdown will exceed the empirical 20th-percentile threshold. Evaluation is strictly out-of-sample via expanding walk-forward cross-validation. We additionally construct a panel of per-stock centrality measures (degree, strength, clustering, eigenvector centrality, PageRank, community) that underpins cross-sectional investment strategies.

1 Data & Target Variable

We use daily returns for the Russell 1000 universe from WRDS (dynamic constituents, mitigating survivorship bias) from 2013 onward. For each date t the forward 21-day maximum drawdown of the index is

$$\mathrm{MDD}_t^H = \min_{s\in[t,t+H]}\left(\frac{W_s}{M_s} - 1\right), \tag{1}$$

where W_s is the cumulative wealth path and M_s its running maximum. The binary crisis label is $y_t = \mathbf{1}\{\mathrm{MDD}_t^H \le q_{0.20}\}$, flagging the worst 20% of drawdown episodes. By construction y_t is strictly forward-looking, precluding any look-ahead bias.

2 Methodology

2.1 Robust Rolling Correlation

At each date t we estimate a $N \times N$ correlation matrix over a 252-day rolling window using exponential weights ($\lambda = 63$-day half-life) combined with Ledoit–Wolf analytical shrinkage [1]. The rescaled return matrix $\mathbf{Z} = \tilde{\mathbf{R}} \odot \sqrt{w \cdot \tau}$ is passed to scikit-learn's `LedoitWolf` estimator; the resulting shrinkage covariance is converted to correlations in the standard way.

2.2 Threshold Graph & Network Features

We threshold $|\hat{\rho}_{ij,t}| \ge 0.75$ to obtain a sparse signed weighted graph G_t. Graph-level features extracted daily include: density, degree/strength moments, weighted clustering, LCC ratio, mean absolute correlation, leading eigenvalue λ_1, and its share of total variance. The degree distribution is heavy-tailed and *stable across market regimes* (Figure 1), as is the assortative degree-mixing pattern and the rich-club structure, suggesting systemic risk manifests through connection intensity rather than topology reorganisation.

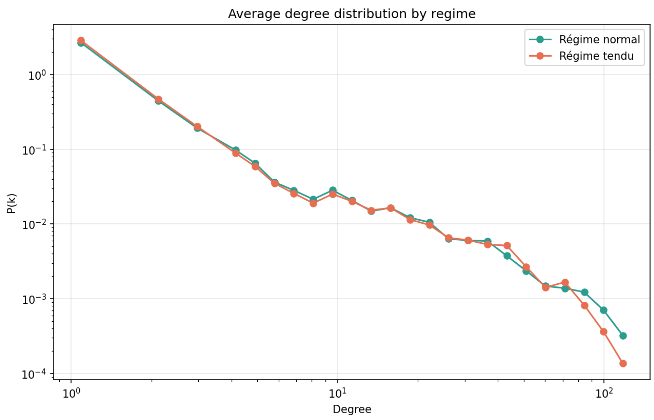

Figure 1: Log-binned degree distribution by regime. The heavy-tailed shape is virtually identical in normal vs. stressed periods.

2.3 Time-Series Features

Over three adaptive horizons (63, 126, 252 days) we compute: compound/mean return, annualised vol and downside vol, drawdown metrics and speed, $\mathrm{VaR}_{5\%}$, $\mathrm{CVaR}_{5\%}$, hit ratio, cross-sectional dispersion, breadth, percentiles, average asset-level vol/drawdown, and cross-horizon ratios/deltas. All features are lag-safe. Total: ≈150–200 time-series columns.

2.4 Walk-Forward Cross-Validation

The scheme uses an expanding training window (minimum 252 days), a 42-day validation block for PR-AUC-based hyperparameter selection, a 21-day buffer, and one OOS prediction per step. Classifiers: L1 Logistic Regression, Random Forest, Gradient Boosting (Table 1).

Table 1: Classification models.

Model	Configuration
LR	$C \in \{0.01, 0.1, 1\}$; SAGA; ℓ_1; balanced
RF	$n = 200$; depth $\in \{5, 10\}$; balanced
GBT	$n = 100$; depth $\in \{5, 10\}$; $\eta \in \{0.05, 0.1\}$

3 Results

3.1 Feature–Target Correlations

Individual Pearson correlations with y_t are modest throughout (Table 2). The most predictive features are medium-term drawdown and mean-return statistics (momentum-in-drawdown effect); the most anti-predictive are share-in-drawdown and short-term volatility metrics (mean-reversion effect). *No network feature reaches either tail of the ranking*, foreshadowing the ablation results.

Table 2: Feature–target Pearson r: 5 most anti-predictive (left) and 5 most predictive (right).

Feature (bottom)	r	**Feature (top)**	r
Share in DD (63d)	−0.17	Market DD (63d)	+0.17
Share in DD (126d)	−0.15	Mean return (63d)	+0.14
Vol ratio (2/252d)	−0.14	Compound ret. (63d)	+0.14
Mkt vol (2d)	−0.13	Market DD (126d)	+0.14
Share in DD (252d)	−0.13	Hit ratio (63d)	+0.12

3.2 Out-of-Sample Predictive Performance

Table 3 reports OOS metrics for all nine (model, feature-mode) combinations. The time-series mode (`ts`) consistently outperforms the network-only mode by ≈0.05 PR-AUC. Combining both feature sets (`all`) does not improve upon `ts` for LR or GBT; RF gains marginally (0.29 vs. 0.28). All PR-AUCs exceed the naive 0.20 baseline, indicating positive but modest lift.

Table 3: OOS classification metrics. Brier: lower is better (naive baseline at 20% base rate: 0.16).

Mode	**Mdl**	**ROC**	**PR**	**Prec**	**Rec**	**Brier**
`ts`	GBT	0.52	0.28	0.30	0.16	0.25
`ts`	LR	0.53	0.27	0.29	0.49	0.29
`ts`	RF	0.52	**0.28**	0.31	0.19	**0.22**
`network`	GBT	0.43	0.23	0.20	0.13	0.29
`network`	LR	0.43	0.22	0.21	0.29	0.27
`network`	RF	0.45	0.23	0.18	0.09	0.24
`all`	GBT	0.50	0.27	0.27	0.14	0.26
`all`	LR	0.53	0.26	0.29	0.52	0.30
`all`	RF	0.52	**0.29**	0.32	0.19	**0.22**

3.3 Market-Timing Backtests

The *oracle* strategy (perfect foresight of y_t) exits to the risk-free asset for every 21-day crisis window and dramatically outperforms buy-and-hold (Figure 2), confirming the economic value of accurate crisis prediction. Prediction-based strategies, however, mostly underperform the benchmark after 10 bps transaction costs (Figure 3): missed crises and false-positive exits from rising markets erode returns. Network-only strategies perform worst; high-recall LR strategies limit drawdowns but sacrifice upside.

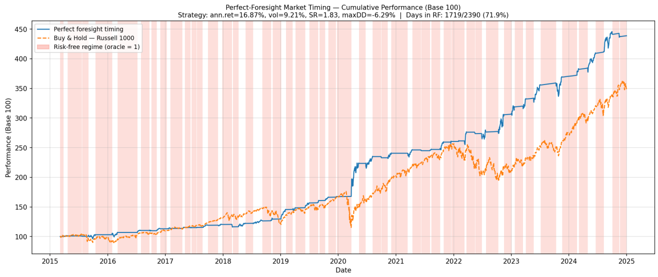

Figure 2: Oracle timing vs. Russell 1000 buy-and-hold. Shaded = risk-free periods. 10 bps/leg.

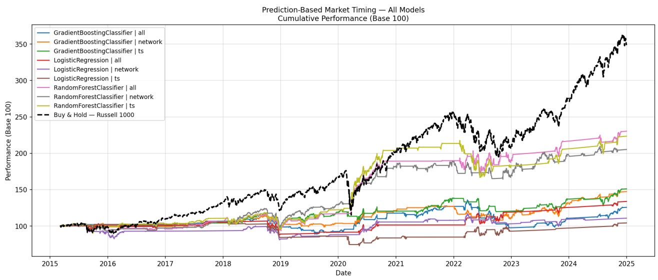

Figure 3: All nine prediction-based timing strategies vs. benchmark (OOS only, 10 bps/leg).

3.4 Cross-Sectional Network Strategies

Node-level metrics (degree, strength, clustering, eigenvector centrality, PageRank, core number) are computed daily and used to form long-only, equal-weighted, daily-rebalanced top/bottom-20% portfolios (Table 4).

The key findings are: **Periphery premium.** Long-bottom portfolios on eigenvector centrality (SR 1.14) and clustering (SR 1.00) beat the benchmark Sharpe (0.89) with volatility ≈14% and max drawdown ≈ −15%—roughly half the benchmark's. This confirms the periphery-premium of [5]: loosely connected stocks carry less co-crash risk and trade at a Sharpe discount. **Central stocks: high return, high risk.** Long-top centrality/clustering portfolios outperform in total return but at proportionally higher vol/drawdown (SR ≈0.95–0.96). **Degree-type degeneracy.** At $\tau = 0.75$, the five bottom portfolios for degree, strength, abs. strength, PageRank, and core number are identical (SR 0.60), as near-isolated nodes have zero values for all these metrics simultaneously.

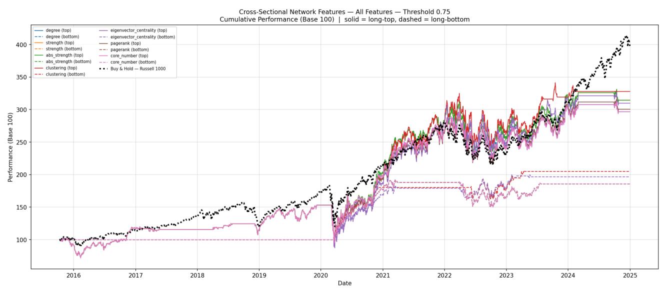

Figure 4: Cross-sectional network strategies. Solid = long-top; dashed = long-bottom.

Table 4: Cross-sectional strategy performance ($\tau = 0.75$), sorted by Sharpe ratio. $\star$=benchmark.

Strategy	**Tot. Ret.**	**Ann. Ret.**	**Vol**	**SR**	**Max DD**
Eigenvec. centrality (bottom)	106.67%	16.37%	14.37%	**1.14**	−15.00%
Clustering (bottom)	90.92%	14.47%	14.48%	**1.00**	−16.11%
Eigenvec. centrality (top)	216.81%	27.22%	28.32%	0.96	−26.60%
Clustering (top)	194.57%	25.33%	26.66%	0.95	−24.87%
Buy & Hold (Russell 1000) $\star$	298.05%	16.17%	18.08%	0.89	−34.58%
Degree/Strength/PR/Core (bottom)	85.83%	6.96%	11.67%	0.60	−22.13%
Strength/Abs. strength (top)	212.90%	13.18%	25.30%	0.52	−35.15%
PageRank / Core / Degree (top)	≈197%	≈12.5%	≈25%	≈0.50	≈ −35%

4 Conclusion

Global network topology features (density, clustering, spectral properties) provide no incremental predictive power for crisis onset beyond time-series return features, with PR-AUC barely above the 0.20 naive baseline. Prediction-based timing does not generate consistent risk-adjusted outperformance after costs. The most actionable finding is the **periphery premium**: daily-updated node-level centrality metrics enable systematic tilts away from systemically important stocks, delivering Sharpe ratios of 1.14 and 1.00 with roughly half the benchmark drawdown. Future work should explore varying the correlation threshold, alternative graph construction (MST, graphical LASSO), and multi-step forecasting.

References

[1] Ledoit, O., & Wolf, M. (2004). A well-conditioned estimator for large-dimensional covariance matrices. *J. Multivariate Analysis*, 88(2), 365–411.

[2] Billio, M., et al. (2012). Econometric measures of connectedness and systemic risk. *J. Financial Economics*, 104(3), 535–559.

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